Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PH507 – SOLID STATE PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Define Miller Indices.
2. Sketch out Diamond structure.
3. What are the imperfections in crystal?
4. What is Line defect?
5. What are the different types of magnetic materials?
6. Give the examples for Diamagnetic materials.
7. What is Polarization?
8. Define dielectric loss.
9. Define Meissner Effect.
10. What is Superconductivity?

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Write a note on Bravais Lattice.

(or)

b) Sketch out NaCl structure.

12. a) Give an account on Laue's method of studying crystal structure.

(or)

b) Discuss Point, line and surface defects.

13. a) Explain the classical theory of Diamagnetism.

(or)

b) Discuss the qualitative explanation of Heisenberg's internal field.

14. a) Write a note on
i) Dielectric constant.
ii) Properties of different types of insulating materials.

(or)

b) Explain the frequency and temperature effects on Polarization.

15. a) Distinguish between Type-I and Type-II superconductors.

(or)

b) Discuss the basic concept of BCS theory.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Discuss Face centred cubic structure and Body centred cubic structure with a neat diagram.

17. Describe the Powder crystal method of studying crystal structure.

18. Give an account on Weiss theory of Paramagnetism.

19. Discuss the various types of Polarizability.

20. Give an account on BCS theory.

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